GW series of MF induction melting furnace

HT- 2.0

User

Manual

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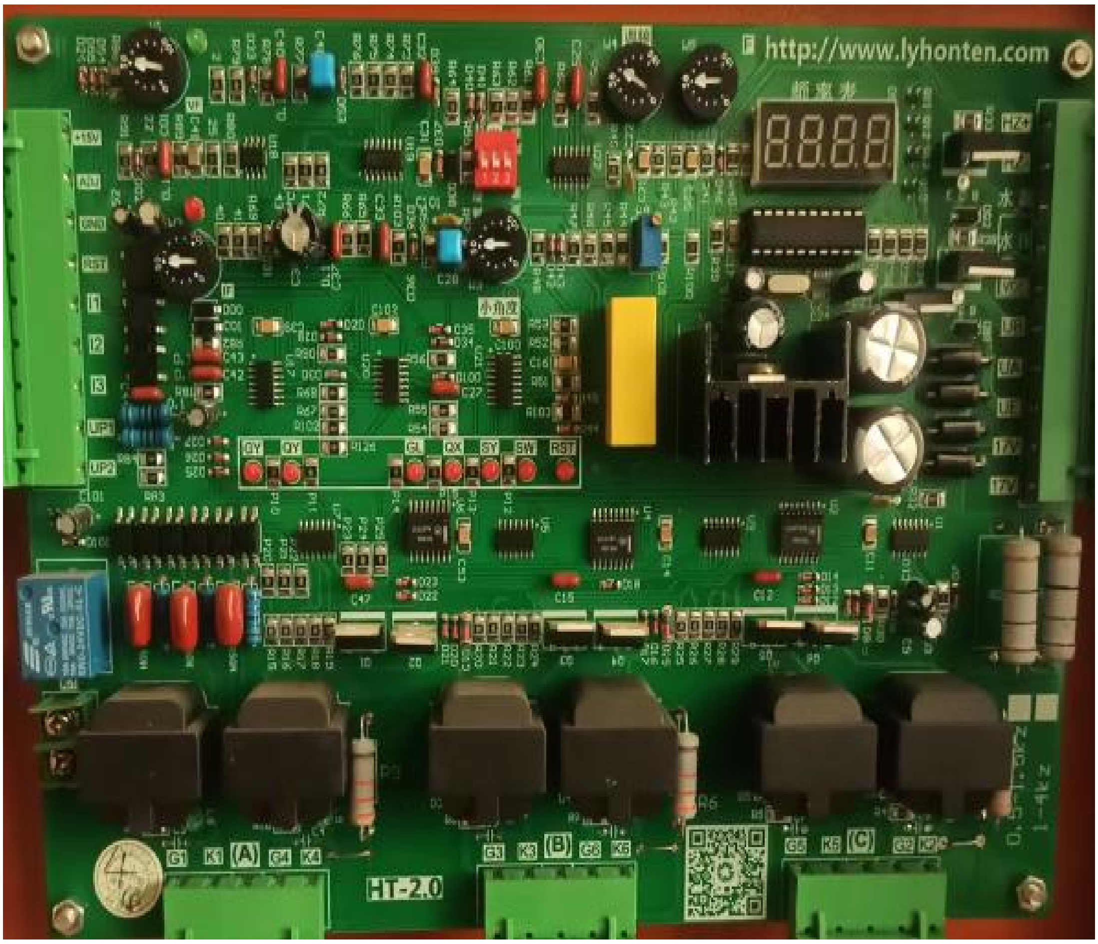


Anti-interference six-pulse control plate

Part 1 Safety Precautions

Safety Definition: In this manual, safety precautions are divided into the following two categories.

Dangers: Dangers caused by failure to operate as required, which may result in serious injury or even death.

Note: Hazards caused by failure to operate as required may result in moderate injury or minor injury, and equipment damage.

Please read this chapter carefully when installing, commissioning and maintaining this system, and be sure to follow the safety precautions required by the contents of this chapter, and any injuries and losses caused by irregularities in operation are not the responsibility of the Company.

1.1 Security matters

Usage Stage	Security Level	
Before installation	Danger	➢ If you find water in the package, missing parts or damaged parts when you open the box, do not install it! ➢ Please do not install the outer packaging logo when it does not match the actual name!
	Attention	➢ Handling should be done gently, otherwise there is a risk of damage to the equipment! ➢ Do not use a damaged drive or a drive with missing parts, there is a risk of injury! ➢ Do not touch the components of the control system with your hands, otherwise there is a risk of electrostatic damage!
Installation	Danger	➢ Please install on metal and other flame retardant objects, away from combustible materials, otherwise it may cause a fire!
	Attention	➢ Do not allow wire tips or screws to fall into the drive or cause damage to the drive! ➢ Please install the drive in a place with little vibration and out of direct sunlight. ➢ When the drive is placed in an enclosed cabinet or confined space, please pay attention to the installation gap to ensure the heat dissipation effect.
Wiring	Danger	➢ The instructions in this manual must be followed and used by professional electrical engineering personnel, otherwise unexpected dangers can occur! ➢ There must be a circuit breaker separating the drive from the power supply, otherwise a fire may occur! ➢ Please make sure the power supply is in zero energy state before wiring, otherwise there is a risk of electric shock! ➢ Please ground the drive properly according to the standard, otherwise there is a risk of electric shock!

	Attenti on	➢ Never connect the input power to the output terminals (**U**, **V**, **W**) of the driver. Pay attention to the markings of the terminals, do not connect the wrong wires! Otherwise the drive will be damaged! ➢ Never connect the braking resistor directly between the + and **-** terminals of the DC bus. Otherwise it will cause a fire! ➢ Please refer to the recommendations in the manual for the wire diameter of the wire used. Otherwise accidents may occur! ➢ Do not remove the connection cables inside the drive as this may cause damage to the drive internals.
Before powering up	Danger	➢ Please confirm that the voltage level of the input power supply is the same as the rated voltage level of the driver; power input terminal (**R**, **S**, **T**) and output terminals (**U**, **V**, **W**) on the wiring position is correct; and pay attention to check whether there is a short circuit in the peripheral circuit connected to the driver, the connected lines are tight, or cause damage to the driver! ➢ Any part of the drive does not need to be tested for voltage resistance, as the product has been tested at the factory. Otherwise, it may lead to Accidents!
	Attenti on	➢ The drive must be covered before it is powered up, otherwise it may cause electric shock! ➢ The wiring of all peripheral accessories must comply with the instructions in this manual and be wired correctly according to the circuit connection method provided in this manual. Failure to do so may cause an accident!
After powering up	Danger	➢ Do not open the cover after power up, otherwise there is a risk of electric shock! ➢ If the indicator light does not light up or the keyboard does not display after power on, please disconnect the power switch immediately and do not touch any input and output terminals of the driver, otherwise there is a risk of electric shock!
	Attenti on	➢ If parameter identification is required, please exclude the possible risk of injury when the motor is rotating! ➢ Do not change the drive manufacturer's parameters at will, otherwise damage to the equipment may result!
Running	Danger	➢ Do not touch the cooling fan, heat sink and discharge resistor to test the temperature, otherwise it may cause burns! ➢ Do not test the signal in operation by non-technical professionals, otherwise it may cause personal injury or equipment damage!
	Attentio n	➢ Avoid dropping anything into the device while the drive is running, as this can cause damage to the device! ➢ Do not use contactor on/off method to control the drive start/stop, otherwise it will cause equipment damage!

When maintaining	Danger	➢ Do not carry out maintenance and repair of the equipment with electricity, otherwise there is a risk of electric shock! ➢ Disconnect the input power for 10 minutes before performing maintenance and repairs on the drive, otherwise the residual charge on the capacitors may cause harm to people! ➢ Do not repair and maintain the drive without professional training, otherwise personal injury or equipment damage will result! ➢ All pluggable plug-ins must be plugged in the case of power failure! ➢ The parameters must be set and checked after replacing the drive.
	Attention	➢ Before performing maintenance work on the drive, make sure the motor is disconnected from the drive to prevent the motor from being damaged by Unexpectedly rotating and feeding back power to the drive.

Part 2 Medium frequency induction melting equipment selection and requirements

1、Main technical parameters of medium frequency melting furnace.

Model Size Specification	Rated capacity (t)	Power Rating (kW)	Rated frequency (kHz)	Rated voltage (V)	Operating temperature (°C)	Melting rate (t/h)	Tilting method
`GWJ-0.1-100/1 (aluminum shell)	0.1	160	1	750	1600	0.14	Electric
GWJ-0.15-160/1 (aluminum shell)	0.15	160	1	750	1600	0.21	Electric
GWJ-0.25-250/1 (aluminum shell)	0.25	250	1	1500	1600	0.35	Electric
GWJ-0.35-250/1 (aluminum shell)	0.5	250	1	1500	1600	0.37	Electric
GWJ-0.5-350/1 (aluminum shell)	0.5	300	1	1500	1600	0.48	Electric
GWJ-0.5-500/1 (aluminum shell)	0.5	500	1	1500	1600	0.72	Electric
GWJ-0.75-500/1 (aluminum shell)	0.5	500	1	1500	1600	0.72	Electric
GWJ-1-500/1 (aluminum shell/steel shell)	1	750	1	1500	1600	0.72	Electric / Hydraulic
GWJ-1-650/1 (aluminum shell/steel shell)	1	750	1	1500	1600	1.07	Electric / Hydraulic
GWJ-1-1000/1 (aluminum shell/steel shell)	1	750	1	2400	1600	1.56	Electric / Hydraulic
GWJ-1.5-800/0.5 (aluminum shell/steel shell)	1.5	800	0.5/1	2400	1600	1.28	Electric / Hydraulic
GWJ-1.5-1000/0.5 (aluminum shell/steel shell)	1.5	1000	0.5	2400	1600	1.56	Electric / Hydraulic

GWJ-1.5-1500/0.5 (aluminum shell/steel shell)	1.5	1500	0.5	2400	1600	2.35	Electric / Hydraulic
GWJ-2-1000/0.5 (aluminum shell/steel shell)	2	1000	0.5	2400	1600	1.56	Electric / Hydraulic
GWJ-2-1500/0.5 (aluminum shell/steel shell)	2	1500	0.5	2400	1600	2.35	Electric / Hydraulic
GWJ-3-1500/0.5 (aluminum shell/steel shell)	3	1500	0.5	2400	1600	2.35	Electric / Hydraulic
GWJ-3-2000/0.3/0.5 (aluminum shell/steel shell)	3	2000	0.3/0.5	3000	1600	3.18	Electric / Hydraulic
GWJ-5-2500/0.3/0.5 (steel shell)	5	2500	0.3/0.5	3000	1600	4.03	Hydraulic
GWJ-7-3000/0.3/0.5 (steel shell)	7	3000	0.3/0.5	4000	1600	4.84	Hydraulic
GWJ-10-5000/0.3/0.5 (steel shell)	10	5000	0.3/0.5	4000	1600	7.45	Hydraulic

2、Equipment operation requirements.

1、Cooling water conditions

a. PH value in the range of 6 to 9.
b. Hardness not greater than 10°G (1°G is equal to 10mg CaO equivalent in 1kg of water).
c. Total solids content ≤250mg/l;
d. The resistivity should be appropriately high, and the values at different operating voltages are

Operating voltage V	Resistivity of water Ω-cm≥
～1000	2000
＞1000～2000	6000
＞2000～3000	14000

e. The secondary cooling water for the IF power supply must be filtered to eliminate all kinds of impurities in the water quality so as not to block the cooling pipes and cause accidents.
f、Inlet water temperature +5～35°C；Outlet water temperature not more than 55°C；Inlet water pressure 0.2～0.4Mpa

2、Power supply conditions

a) Three-phase voltage unbalance ≤5%
b) Grid voltage fluctuation ≤±10%
c) Grid voltage waveform distortion rate ≤10%
d) Control system supply voltage ≤5%
e) **Note: The equipment should be powered by an independent transformer is better, as far as possible, not mixed with other equipment power. The equipment full power operation power factor ≥ 0.95, can not add**

compensation device; such as mixed power with other equipment, need to add compensation device, because the three-phase controlled voltage regulation will inevitably bring harmonic interference, it is possible (depending on the circuit distribution inductance) between the compensation device and the medium frequency power supply to produce a superposition of high harmonics, affecting the normal input of the compensation device.

3、Geographical environment

a) Altitude ＜3000m

b) Ambient temperature 0-42℃

c) Relative temperature ＜90% (average temperature not less than 20℃)

d) No surrounding conductive dust, explosive gases and corrosive gases that seriously damage metals and insulation

e) No significant vibration and bumps

f) Installation method: Indoor

Part 3 Introduction to Medium Frequency Power Supplies

1、Mid-frequency power supply electric cabinet: adopt the upper in and lower out line. The arrangement of the cabinet is simple and reasonable, the main circuit is isolated from the control circuit, the wearing parts are easy to replace, and there are necessary safety protection measures.

2, the **main circuit**: the power supply uses silicon controlled parallel resonant circuit, which has high reliability, easy protection, and the inverter bridge has a certain fault tolerance for the trigger pulse, compared to the series resonant circuit and IGBT power supply is easy to achieve short-circuit protection, is the preferred power supply structure of large foreign companies.

The inverter SCR is made of national brand, the reserve number of current and voltage is ≥2.5 times, the rectifier element is made of national brand, all SCRs are subjected to severe 72 hours continuous high temperature blocking and full dynamic testing, so that the installed components have high reliability; the external absorption parameters of the components are instantaneously and dynamically optimized for screening, so that the irreversible deterioration caused by the phase change spike voltage on the components is reduced to a minimum. fundamentally improves the service life of the equipment.

3、Compensation capacitor: large-capacity electric heating capacitor, so that the complete set of equipment covers an area as small as possible.

4、Control circuit.

(1) Adopt all-digital ultra-large-scale integrated circuit, single-board structure, and all devices are screened by high-temperature aging. The control circuit adopts digital control and has strong anti-interference ability. In addition to the conventional control functions of rectifier, inverter, overvoltage, short circuit, current limit and voltage limit, it also has fault self-diagnosis function and fault delay function.

(2) Adopt zero-voltage intelligent scanning starting circuit to ensure the reliability of starting with crucible pot mode and starting with full and freezing furnace; meanwhile, it has conventional automatic repeat starting function.

(3) The output power is continuously adjustable to meet the needs of oven, melting, heating and keeping warm.

(4) The specially designed inverter angle control loop can adjust the equivalent load impedance in real time by controlling the inverter angle according to the working condition of the load, so that the power supply and the load are in the best match, so that the power supply is always in the maximum possible output state, and the power factor can reach above 0.95, which maximizes the output of the equipment. In this way, the equipment can reach constant power output within the whole range of working conditions allowed by the process, which speeds up the melting speed; after reaching the rated temperature, it can realize automatic heat preservation and save power.

(5) Only one potentiometer is needed to adjust the output power, which realizes "foolproof" operation and avoids damage to the equipment due to misoperation. In the production process, there is no need to manned the power status and other operations except power setting.

(6) The voltage and current are controlled by double closed-loop non-differential control, which not only can overcome the fluctuation of the power grid and the disturbance of the load change, but also is non-differential regulation in the voltage and current limiting condition, without the loss of equipment output caused by the traditional voltage and current cut-off conditions.

5, anti-interference 6 pulse control board manual

KGPS-6D IF power supply control board is a new type of six pulse wave control board developed by our company for IF power supply with 3-phase rectification in the main circuit.

KGPS-6D IF power supply control board mainly consists of power supply, regulator, phase shifting control circuit, protection circuit, start-up algorithm circuit, inverter frequency tracking, inverter pulse formation, pulse amplification and rectifier pulse transformer. The circuit is digitalized except for the regulator, so it has the characteristics of high reliability, high pulse symmetry, strong anti-interference ability and fast response time.

KGPS-6D IF power supply control board and the previous control board have made a lot of improvements in protection, after all the protection actions, it needs to be manually reset to start, which can avoid the power supply starting by itself after the fault disappears; such as water failure, phase loss. At the same time, it is convenient for maintenance personnel to find faults. This control board has a function to detect whether the power potentiometer back to zero, after the power supply to power and reset automatically detect the potential of the potentiometer ADJ end, if the ADJ end is not zero potential will start a forced reset, RST indicator light, blocking the rectifier pulse, the potentiometer back to zero, RST automatically extinguished, rectifier pulse open, you can start normally. This function can avoid the damage to the equipment and the impact to the power grid caused by the wrong operation of high power start, thus reducing the failure rate of the power supply and the danger to the staff.

The control board is available in two types, the fence terminal (KGPS-6D) (KGPS-6C) and the plug-in terminal (KGPS-6DC), and there are two types of sweep frequency ranges: 0.5-1.5 kHz and 1-4 kHz.

Fault Indicator

Code Name	Indicates status when light-emitting diode is on
GY	Medium frequency overvoltage fault
QY	Power supply voltage low fault
GL	Rectifier bridge overcurrent fault
QX	Rectifier bridge out of phase fault
SY	Low water pressure failure
SW	High water temperature fault
RST	Reset switch is closed or reset after the power potentiometer does not return to zero

Function	Connector	Description
Voltage feedback signal	UP1 UP2	Medium frequency voltage transformer 20V (voltage limit 8-20v)
Current feedback signal 1	I1 I2 I3	Rectifier bridge current transformer AC three-phase 10V (7-14v)
Given Reset	+15V	+15V power output for power potentiometer
	ADJ	Power potentiometer 0-15V
	GND	Power potentiometer ground terminal, and reset common
	RST	Hover for operation, connect to GND for stop operation and fault reset
Power supply	17V	AC 17V 50W
	17V	

Inverter pulse output	UE	Inverter output common +22V	
	UA	Inverter pulse output A	
	UB	Inverter pulse output B	
Water temperature protection	Water temperature	Connect the water temperature gauge contacts, normally open when working, and 15V closed when the water temperature is high	
Water pressure protection	Water pressure	Connect the water pressure gauge contacts, normally open when working, when the water pressure is low and 15v Closure	
External frequency meter	HZ+	External frequency meter + terminal	
	HZ-	External frequency meter - terminal 1mA or 5mA	
Rectification pulse signal output	G1	No. 1 thyristor control pole	Note that this signal has been modulated into a pulse string, connected to the SCR control pole and cathode
	K1	No. 1 thyristor cathode	
	G4	No. 4 thyristor control pole	
	K4	No. 4 thyristor cathode	
	G3	No. 3 thyristor control pole	
	K3	No. 3 thyristor cathode	
	G6	No. 6 thyristor control pole	
	K6	No. 6 thyristor cathode	
	G5	No. 5 thyristor control pole	
	K5	No. 5 thyristor cathode	
	G2	No. 2 thyristor control pole	
	K2	No. 2 thyristor cathode	

Switch function	Serial number	Description
Repeat start switch	1	ON OFF Repeated start, for detecting whether the DC voltage is normal, when working In off
Inverter angle automatic adjustment switch	2	ON Inverter angle is automatically adjusted, OFF Operates only with a small angle and works at ON
High Frequency Low Frequency	3	ON switch closed, low sweep frequency rate, off sweep frequency rate high, change range 1 times

Potentiometer function and use

W1 (VF) maximum IF voltage setting potentiometer (voltage limit) clockwise small, maximum adjustment range approx. 3 times. Overvoltage values are

1.2 times the pressure limit value.

W2 (IF) rectifier bridge maximum output current setting potentiometer (current limit) clockwise small, maximum adjustment range of about 2 times. Over

The current does not need to be adjusted, when the current limit value is adjusted, the overcurrent value automatically becomes 1.4 times the current limit value.

W3 (small angle) minimum inversion lead front angle setting potentiometer, clockwise large, maximum adjustment range of about 20 ° - 40 °

(factory setting 30°).

W4 (frequency) maximum it excitation inverter frequency setting potentiometer, clockwise for increasing, the maximum adjustment range of about 2 times

W5 (F) external frequency meter calibration potentiometer, clockwise for increasing readings.

Commissioning of MF cabinet

Rectification part commissioning

For the safety of commissioning, the rectifier bridge should be commissioned separately first. Therefore, before commissioning, disconnect the reactor copper row, or disconnect the inverter pulse signal to make the inverter quit working. Connect a dummy load (you can use several light bulbs in series) at both ends of the positive and negative stages of the rectifier bridge. Turn the control board switch 1 to ON position, (so as to turn off the repeated start); use an oscilloscope to prepare for measuring the output DC voltage waveform of the rectifier bridge; turn the power potentiometer counterclockwise to the minimum.

Close the control power switch, send the main power supply, press reset, check if there is a fault indication such as phase loss protection, if there is a fault as indicated first troubleshoot. Turn the power potentiometer clockwise large, should be able to on the oscilloscope. See the rectifier bridge output waveform, from the minimum gradually up to the full open, and 6 wavehead complete. Then turn the power potentiometer to the minimum, the rectifier bridge waveform should drop to the minimum, disconnect the main power after debugging.

Inverter section commissioning

W3 is a small angle potentiometer. W3 is a small angle potentiometer. The small angle is 30°, the large angle is set to 60° at the factory, generally no need to adjust, W5 is an external frequency meter calibration potentiometer, calibrate it against the digital frequency meter.

Install the reactor copper strip, or turn on the inverter pulse signal to put the inverter into operation and check if the inverter pulse signal is normal with an oscilloscope. Turn switch 1 to the off position to turn on the repeat start. If the main power supply is turned on, adjust the power potentiometer to make the IF start, if it cannot start, adjust the voltage feedback signal to 20V, check whether the inverter SCR voltage is normal after starting, then the power can continue to increase until the maximum, and look at the DC current and IF voltage values. Wait for the voltage or current to change before adjusting, otherwise the adjusted value is not accurate. Adjust W1 IF voltage to work at rated value, adjust W2 to make DC current reach rated value.

If you have any questions, please call our company for consultation. Good luck with your commissioning!

Statement

Due to the large power of the IF power supply, this control in our company can only do analog signal test, if found in the debugging problems, please contact the company to return, any device damage outside the control board our company will not be negative.

Medium Frequency Power Supply Operation Instructions and Maintenance Manual

I. Operating instructions

1. Panel arrangement (see wiring diagram for details)

2. Preparation before opening the furnace

1) Checking for loose fasteners in various parts.
2) Checking whether the insulation of the main power supply in the cabinet is good.
3) Checking whether the water pressure, flow rate and temperature of each water circuit are normal.
4) Checking the normal working instructions on the "central control panel".
5) Checking whether the trigger pulse (rectifier, inverter) indicator is normal.
6) Check whether the induction coil and capacitor insulation of the furnace body are normal.
7) Checking water-cooled cables for reliable contact.
8) Check whether the voltage of each incoming line is normal

All of the above are normal conditions under which the IF power supply can be started.

3. Power supply operation (Non-professional operators are not allowed to turn on the IF power supply)

3.1. Start-up operation

（1） Closing the air switch Q1 in the cabinet to supply power to the secondary circuit.

（2） Starting pumps to supply water to the cooling system.

（3） Press the "Control Power Close" button to power up the control power supply.

（4） Pressing the "mains power on" button to power up the main circuit.

（5） Right turn the "start/stop" button to prepare for IF power start.

（6） Slowly adjust the "power adjustment" potentiometer PR and pay attention to the frequency table, if there are indications and can hear the medium frequency call, indicating that the start-up success, after the successful start-up potentiometer PR once rotated to the bottom, while the main control board on the "start P.P" light goes out, the "voltage ring V.LOP" light. "LOP" light is on. If the start is unsuccessful, it needs to be restarted.

3.2. Stop operation

（1） Turn the power adjustment potentiometer PR counterclockwise to the end and all indicator meters to zero.

（2） Press the "Start/Stop" button to stop the IF power supply.

（3） Press the "main switch" button to disconnect the main circuit.

（4） Press the "Control Power Sub" button to control the power off.

（5） Disconnecting the air switch Q1 in the cabinet.

4. Other notes

1) When the fault stops, the "central control board" can keep the memory, and the power can be restarted only after the fault is removed and the IF "start/stop" button SA1 is pressed.

6) In case of failure or emergency, press the IF "start/stop" button first, and then stop the power supply according to the power supply stop procedure, and then restart the power supply only after the failure is removed.

7) The pump stopping time should be decided according to the water temperature in the inductor, generally the water supply should be stopped about 240 minutes after the power supply is stopped, the cooling water of the power supply can be stopped immediately.

Second, the fault and alarm

The device is set up with a set of relatively complete fault and alarm system, when the system fault occurs, the indicator light related to it lights up and points out the type of fault, the personnel concerned can judge the cause of the fault according to the indicator light and take corresponding measures.

1. MF power main control board failure

There are five types of faults in the main control board: overcurrent, overvoltage, phase failure, water failure, and undervoltage.

1.1. Overcurrent fault

The following phenomena occur during overcurrent.

a. The power failure light on the power cabinet is on.

b. IF power pull inverter (α=150 degree), in shutdown state.

c. "Overcurrent **GL**" light-emitting diode on the main core control board is on.

1.2. Over-voltage fault

The following phenomena occur during overpressure:

a. The power failure light on the power cabinet is on.

b. IF power pull inverter (α=150 degree), in shutdown state.

c. The "Over Voltage **GY"** light tube on the main control board is on.

1.3. Phase loss fault

The following phenomena occur when the phase is missing.

a. The power failure light on the power cabinet is on.

b. IF power pull inverter (α=150 degree), in shutdown state.

c. The "Out of phase **QX**" light tube on the main core control board is on.

1.4. Water shortage failure

The following phenomena occur when water is scarce.

a. The power failure light on the power cabinet is on.

b. IF power pull inverter (α=150 degrees), in shutdown.

c. The "water pressure **SY**" light-emitting tube on the main control board is on.

1.5. Undervoltage fault

The following phenomena occur when undervoltage is present.

a. The power failure light on the power cabinet is on.

b. IF power pull inverter (α=150 degrees), in shutdown.

c. The "undervoltage **QY**" light tube on the main control board is on.

Third, the medium frequency power supply maintenance and maintenance

However, the overload capacity of the semiconductor device is poor, therefore, reasonable use, correct operation and careful maintenance are important guarantees for the safe operation of the thyristor inverter power supply to avoid failure. In the continuous operation of the production line, it is particularly important to improve the maintenance of the device.

1, often clear the accumulated dust inside the distribution cabinet, especially the outside of the thyristor core, to wipe clean with alcohol. The frequency conversion device in operation is usually a special room, but the actual operating environment is not ideal. During melting, there is a lot of dust and strong vibration. Therefore, attention must be paid to frequent cleaning to prevent malfunctions.

2, often check whether the water pipe joints are firmly tied, clean the cooling water pipe inside the scale and blockage. When the water pipe is aging and cracked, it should be replaced in time. When the device is running in summer, the outside of the thyristor core is easy to condensation, causing serious leakage between the thyristor anode and cathode, should consider reducing the temperature difference between the inlet and outlet water, and should stop running when the condensation is serious.

3、Regularly overhaul the device, check and tighten the screws, nuts and crimp fasteners of each part of the device; circuit breakers, contactors and relays with loose contacts and poor contacts should be repaired and replaced in time, do not use reluctantly to avoid causing greater accidents. Regularly calibrate the device's over-current, over-voltage, current-limiting, voltage-limiting calibration values, check the reliability of the action of the protection system, to prevent the failure of protection.

4, often check whether the wiring of the load is intact and whether the insulation is reliable. Melting furnace in the replacement of new furnace lining, should pay attention to check the insulation. Medium frequency power supply device work site environment is poor, the failure rate is relatively high, and is often ignored. Strengthen the maintenance of the equipment to prevent the furnace inductor failure wave and power supply, is an important part of ensuring the safe operation of the equipment.

Four, in the frequency cabinet common troubleshooting

The initial adjustment of the power supply or the failure of the power supply during operation, the failure of the whole machine to start, accompanied by certain phenomena, are described as follows.

1. After the control switch is closed, the indicator light on the control board does not light up. Possible causes of this fault are.

    1.1 Control switch is bad, use a stylus or multimeter to measure whether the incoming and outgoing end of the control switch is charged or whether the voltage is normal, if not, confirm the cause of the fault and take appropriate measures to make its voltage normal.

    1.2 bad power transformer or wiring is not secure, first tighten the wiring, if still intact, you can disconnect one terminal of the transformer, use a multimeter to measure the transformer coil on and off, if the coil is broken, then the power transformer needs to be replaced.

    1.3 Control the power fuse is broken, can be measured with a stylus, confirm the insurance can be replaced.

2. The main circuit air switch does not close.

    2.1 The water pressure is low and the water pressure relay is not connected, so you should try to make the water pressure normal.

2.2 The air switch loss-of-voltage protection coil is not energized. Check the loss-of-voltage protection circuit, remove the fault in the line, and energize the coil.

2.3 The air switch will jump when it is closed. There may be a short circuit in the main circuit, should be carefully checked, so that the short circuit fault eliminated, and then send power to work. Or because the air switch is not the right setting of the detent current, should be re-adjusted.

3. Static DC voltage is not normal.

3.1 Low AC input voltage.

3.2 rectifier output is out of phase, check whether the control board output pulse is normal, the SCR control pole and cathode wiring is reliable.

3.3 Poor contact at each connection point of the main circuit is good.

3.4 Rectifier SCR bad.

4. The device does not start.

4.1 The inverter pulse is not normal.

4.2 The inverter check switch is not set to the self-exciting gear (working position).

4.3 The current transformer on the IF capacitor is bad or miswired. Replace the transformer terminals or measure the transformer coil on-off condition.

4.4 The voltage signal on the IF transformer is wired incorrectly.

4.5 Voltage or current signal potentiometer is damaged or wired unreliably, fix the wiring or check the potentiometer and repair.

4.6 Whether the incoming resistor (5 W, 470 - 600 Ω) on the control board is damaged.

4.7 Check the complete line of voltage and current integrated signal for errors.

4.8 The IF capacitor has a short circuit or is damaged.

4.9 The induction loop has a turn-to-turn short circuit or a short circuit to ground.

4.10 Inverter bridge circuit is faulty.

5. Static DC voltage is normal, but the voltage cannot rise after adding load.

5.1 The current limit adjustment value is too low, at this time the equipment should be "buzzing" abnormal sound, you can adjust the current limit adjustment potentiometer, so that it works normally.

5.2 The pressure limit adjustment value is too low, adjust as above.

5.3 Inverter bridge in a silicon controlled breakdown or not working. Use an oscilloscope to observe the trigger pulse and the waveform at both ends of the SCR, so as to confirm the corresponding measures to eliminate the fault.

5.4 Failure of other parts in the inverter bridge.

6. The device works normally, the voltage suddenly decreases, and there is " buzz " ground abnormal sound.

6.1 rectifier bridge has a silicon controlled breakdown or does not work.

6.2 The inverter bridge has a silicon controlled breakdown or does not work.

6.3 One of the rectifier bridge is damaged by fast fusion.

6.4 Pressure or current limiting function caused by.

7. After the overvoltage is adjusted, the overvoltage protection will be activated by adjusting the voltage limiting potentiometer.

This is a normal phenomenon, should be adjusted to the appropriate position of the voltage limiting potentiometer, according to the debugging instructions to readjust the overvoltage rectification value, so that it works stably.

8. The voltage rises to the maximum during operation, but the current is low and the power is also low.

8.1 Shunt resonant capacitor capacitance is small, the capacitor should be increased.

8.2 The number of turns of the induction loop can be changed to vary the current.

8.3 The thickness of the furnace liner can be changed to vary the current.

9. Static DC voltage is not capable of full conduction.

9.1 Phase sequence error, adjust any two-phase line can be.

9.2 Adjust the power potentiometer resistance value is not correct.

9.3 The protection circuit is always in operation.

10. Press the IF start button, adjust the power potentiometer, the power supply does not respond or only DC voltage without IF voltage, the reason may be.

10.1 The load is open, i.e., the sensor is not connected.

10.2 Inverter pulse power is too small or no pulse, and the inverter tube is not triggered.

10.3 The rectifier circuit is faulty and there is no rectifier output.

11. After pressing the IF start button, the overcurrent protection operates and the rectifier is pulled into the inverter state. For the newly installed power supply, check whether the voltage sampling polarity is correct, whether the polarity of the inverter pulse is correct, and whether the leading angle is too small. For the polarity problem that does not exist in the already operating power supply, it can be analyzed from the following aspects.

11.1 Thyristor is damaged or not, measured with a multimeter.

11.2 Whether the fuse is broken, if broken replace it.

11.3 Whether the load circuit is short-circuited and the load is too heavy. Observe the load oscillation waveform with an oscilloscope to determine.

11.4 Start the lead angle is too small, appropriate adjustment of the lead angle.

11.5 Whether there is interference with the inverter pulse.

11.6 Whether the overcurrent adjustment value has changed and needs to be re-adjusted.

11.7 Whether the current feedback is too large, the feedback is too large also makes the oscillation stop.

11.8 The rectifier circuit is faulty and the DC output is too low.

11.9 Whether the insulation of the IF power supply is reduced.

11.10 Whether the voltage feedback signal is disconnected.

In short, to troubleshoot the power supply as soon as possible, not only requires a strong technical level of maintenance personnel, but also requires a wealth of experience, but also requires the use of manufacturers in the usual maintenance and repair of the power supply, in order to ensure the normal operation of the power supply.

Part 4 Introduction to Medium Frequency Induction Furnace

I. Structure Introduction

1. Furnace body part

The mechanical part of IF furnace is composed of furnace body, water and electricity introduction system, tilting furnace device, etc.

1.1 Furnace body

The furnace body includes furnace shell, inductor, etc.

1.1.1 Furnace shell

It has the advantages of good rigidity, compact structure, easy maintenance, etc. Its rotary hole is machined at one time to ensure concentricity, so that the furnace shell and rotary shaft will not be deformed during the tilting process due to different centers.

1.1.2 Sensor

The inductor is the heart of the furnace, which not only directly affects the absorption of the furnace power, but also the rationality of its design and manufacturing quality directly affects the service life of the furnace lining. The inductors produced by us have the following characteristics.

a) The inductor parameters are optimized by the computer to obtain a set of the most satisfactory technical parameters for the designer to carry out the structural design.

b) The inductors are made of thick-walled high-quality T_2 purple copper tubes, which are wound on special winding equipment to ensure the manufacturing quality and the flatness of the inner wall of the coils, further improving the service life of the furnace body.

c) The inductor turn spacing is fixed by a sufficient number of epoxy posts to make the induction coils integral, thus reducing vibration and noise during operation.

1.2 Water and electricity introduction system.

The furnace water circuits are connected in parallel with each other, usually with open softened water cooling. (See water system schematic)

1.2.1 The introduction of water and electricity are used after the form of line out, reducing the line loss and improving the service life of water-cooled cables.

1.2.2 The water system should be equipped with a pressure sensor on the total inlet pipe in order to adjust the water flow of each cooling branch to ensure the normal operation of the equipment.

1.2.3 Water-cooled cables are crimped to copper stranded wire using a cold compression molding process. This method of connection is strong, low contact resistance, does not damage the copper strand and can be quickly replaced, and can withstand 0.5Mpa of water pressure without leakage or rupture.

1.3 Tilting furnace system

The tilting mechanism adopts worm gear reducer (electric, manual), or hydraulic tilting.

Part 5 Introduction to Furnace Lining Fabrication

1. The installation of this system, although not very complicated, is very demanding. This is because the workplace is special - strong electrical energy, water flow, high temperature molten metal and oil pressure system are working in a very compact contradictory environment, especially the quality of the furnace lining manufacturing, maintenance level and operator's responsibility will affect the life of the furnace lining, and even the occurrence of leakage, coupled with improper operation, in serious cases, the Will also cause metal spattering and explosion accidents. Therefore, we must pay great attention to, here only on the problem of the lining prone to problems to explain. Furnace lining must be able to withstand high temperatures, high temperature mechanical strength, good thermochemical stability and electrical insulation properties. In addition, there must also be a small coefficient of thermal expansion, resistance to severe cold and heat performance is good. (Coil refractory mastic must be sufficiently dry)

2, cast iron melting commonly used quartz sand as furnace lining, according to a certain particle size ratio, add the appropriate amount of boric acid as a binder, and then stir well can be used. The quartz sand requires less alkaline impurities, less metal oxides, high quartz content, and its composition is as follows.

SiO_2 98.7~99.3%

Al_2O_3 0.3 to 0.5%

Fe_2O_3 0.01 to 0.3%

TiO_2 0.05~0.15%

Quartz sand should be free of inclusions, and by hand picking and magnetic separation, sieved and dried at about 120 ℃ for 5 ~ 6h to remove moisture (generally about 0.5%).

Quartz sand particle size ratio: coarse particles play a skeleton role in the furnace lining, increasing the strength of the lining; fine particles play a filling and bonding role. In order to obtain a higher dense furnace lining, according to the production practice, the following formula is recommended for reference.

6 -8 mesh: 30%; 10-20 mesh: 20%; 40-70 mesh: 15%; 70-140 mesh: 15%; 200-270 mesh: 20%

External industrial boric acid 1.2 ~ 1.8%, the higher the working temperature the smaller the amount of boric acid, the general ratio: bottom 0.5 ~ 1.0%; furnace wall 1.5 ~ 1.8%; furnace mouth 1.8 ~ 2.0%.

3、 The furnace lining structure is generally shown below, the laying order is from outside to inside one by one (see the "furnace body" drawing provided).

Each layer of joints at the lap width of 30 ~ 40mm, should be laid close to the outer layer, and then take turns with "up ring" fixed, not to make the slide off.

4, the knotted material must be stirred evenly to prevent impurities mixed in and secondary moisture absorption. In recent years, the domestic has been able to provide a variety of quartz sand mix, sealed bag packaging, can be more long-term preservation, the user can directly use this furnace material, without further mixing, and the material contains boric acid, shorten the baking time.

5、Furnace construction tools.

There are two types of furnace building tools: manual and electric.

The former needs to be homemade, commonly used manual furnace building tools such as steel fork, flat hammer. : The

The "manual - electric" furnace building machine, with a steel fork and flat hammer larger than the above picture, will be connected to the electric vibration machine head and steel pipe, 1-2 people to operate, more than twice the efficiency of the manual.

There are also some mechanized large furnace building machine "electric, pneumatic", more efficient, as long as the preparation work is adequate, all the furnace building time in 0.5 ~ 1.5 hours all completed, and the texture of uniform.

6、 Manual furnace construction.

6.1 Furnace bottom: first, the "first electrode" bundle, after leaving enough length, the rest of the penetration into the furnace bottom hole and asbestos rope plugging, with mortar filling seal. The bottom of the furnace in layers, fill sand pounding, the first layer of 100mm thick, after each layer of 60 ~ 80mm, pounding with a steel fork, by 4 ~ 5 people at the same time. As each person pounding differences, it is required while pounding, while moving around the mouth of the furnace, continuous pounding 5 ~ 6 times and then pounding with a flat hammer, and then pick up loose surface layer, continue to fill the next layer of sand, so that it is not easy to do layering.

6.2 Furnace wall: When the bottom of the furnace is beaten to 20-30 mm higher than the predetermined thickness, shovel off and scrape the excess, correct the level of the bottom with a level, check to the size of the top of the furnace, pound it again with a flat hammer, cut off the first electrode exposed by 50 mm and put it into the crucible steel mold. Find the right center and fix it with a wooden wedge (the crucible mold can be divided into two sections for more than 7-10 tons to facilitate manual knotting) and start knotting the furnace wall. Because the wall is much thinner than the bottom of the furnace, it is subjected to large temperature changes and violent scouring of the iron, so it has to be pounded especially densely. Furnace mouth parts more than 300mm plus another water glass solution (1:1) 5 ~ 6%. Each layer knotting method as above.

7、 Oven and sintering.

Oven drying and sintering of the furnace lining is an important part of obtaining excellent high temperature strength. The sintering process is determined by is the polycrystalline conversion properties of the quartz sand and the capacity of the furnace.

7.1 In order for the user to develop the furnace lining baking and sintering process, the principles of transformation and heating rate during the baking process of the "quartz sand-boric acid" system are given here for reference.

Temperature ℃	Internal changes in the furnace lining	Heating up speed principle
<500	Mainly to remove water, including water of crystallization released from boric acid to boric anhydride	Loose furnace lining, water vapor is easy to penetrate, but the furnace lining around to prevent steam from escaping, the early speed can be faster, at about 400 ℃ insulation exhaust 1h
500～650	Boric acid starts to change, low temperature quartz starts to transform and liquid phase appears at the perimeter	Prevent the evaporation of boric acid transfer, should accelerate the speed of temperature rise
650～850	No quartz transformation	Consider the furnace lining uniform temperature and continue to rapidly heat up
850 to 1250	Quartz begins to shift to lepidocrystals but not drastically, entering preliminary sintering	Slowing down the heating rate
＞1250	The quartz is intensely transformed to lepidocrystals, and then to square quartz at 1470°C	Expansion and cracking tendency is very large, should be slow heating, in 1500 ℃ ~ 1550 ℃ insulation 2 ~ 3h

It must be noted: SiO_2 polycrystalline transformation is very slow, even if the oven sintering is completed, that is also the surface layer is very thin layer, after thickening to 10 ~ 30mm, which is related to the furnace temperature. Near the inductor for the loose layer, by the iron for the sintering layer, the middle for the transition layer. This structural characteristic can prevent the cracking through the burner, can maintain the integrity and reliability of the furnace lining, and is very beneficial to continuously promote the densification of the furnace lining, which is the advantage of other general refractory materials do not have.

The main points of baking and sintering are: low temperature slowly rises, high temperature full furnace sintering, furnace material requires low carbon and less rust. The whole process is carried out in three stages.

7.2 Baking stage: Before the crucible mold reaches 1100°C, it must be heated slowly so that the crystalline water in the boric acid (about 43% by weight) and the moisture in the lining material are slowly drained out of the furnace. In addition, also because quartz has polycrystalline transformation at 573°C, 867°C and 1025°C, causing volume expansion, slow heating in this interval can make the local expansion stress spread out and less cracks, so the heating rate is 100~140°C/h, which takes 9~12h.

7.3 Melting stage: When the temperature of crucible mold reaches 1100~1200°C, add small pieces of material one after another, in order to reduce the fluctuation of furnace temperature too much, each feeding should not be too much, about 200~400kg is appropriate, when all melting makes the iron surface rise to about 250mm at the furnace mouth. This stage of the early (not melted) power is high, about 60% of the rated power, can make the furnace material quickly warm up, reduce the temperature difference between the furnace lining. To be fully melted before, should immediately reduce the power to reduce the scouring force to continue to heat up, taking about 5h or more.

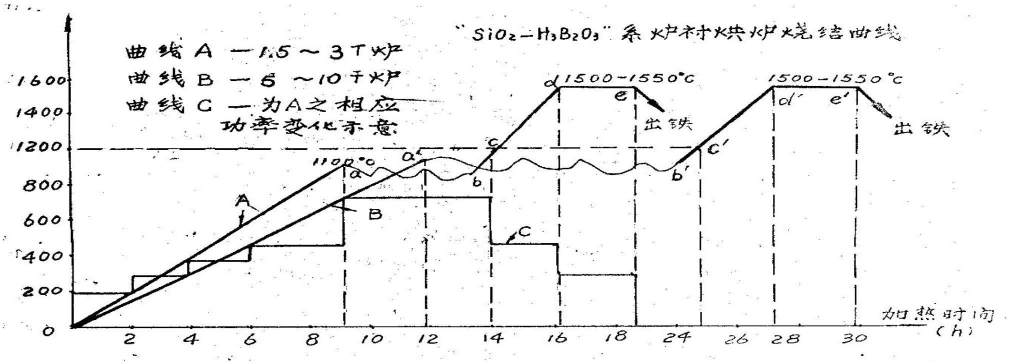
7.4 sintering stage: continue to heat up to about 1550 °C after holding 2 ~ 3h, in order to make the furnace lining at high temperature into a hard shell with sufficient strength, and then pour out all the iron, check the surface sintering condition of the furnace lining, in the return of 1/2 furnace iron, add material continuous melting at least 2 ~ 3 furnace before stopping the furnace. In this stage, because the furnace lining still contains water vapor to prevent the impact on the induction ring insulation, the lining strength is poor, to reduce the harm of metal liquid stirring and scouring, so the power feed does not exceed 75 ~ 80% of the rated power, keep in mind.

7.5 insulation, cooling: furnace insulation or empty furnace cooling, should be covered with a good cover, sealed tightly around, to reduce the amount of water supply, so that the furnace lining slowly cooling to achieve uniform shrinkage, reduce cracks.

7.6 Sintering furnace lining with low temperature iron from other furnaces: In plants with conditions, in order to reduce the repeated charging temperature drop, the crucible steel mold can be filled with iron (iron from other furnaces) before melting and heated up with low power, other as before.

Note: The thermocouple should be placed against the crucible mold at furnace temperatures <500°C and then moved to the center of the furnace chamber. Also, the baking curve is achieved by adjusting the power and intermittent power feeding both together.

7.7 Oven sintering reference curve: According to the above principles and practice, the following oven curve is developed for reference.



Note: Curve A is suitable for 1.5 ~ 3t furnace; curve B is suitable for 5 ~ 10t furnace, other furnace models can refer to this principle to develop. Here the factors affecting the length of time depends mainly on the exclusion of lining water fraction and successive charging capacity size. For example, the total time of 20 tons furnace in more than 60h.

The warming section on the curve 0-a.

a-b on the curve is the repeated dosing section.

The heated melting section on the curve b-c.

c - d on the curve for the superheated section.

The curve d-e is the holding preliminary sintering section.

7.8 Recommended dimensions of the crucible die (reference).

Furnace tonnage (ton)	0.1	0.15	0.25	0.35	0.5	0.75	1.0	1.5	2.0	2.5	3.0	5.0
Inductor inner	360	380	420	480	580	620	720	760	820	860	920	1140

diameter mm												
Average crucible diameter mm	220	240	280	330	420	460	540	550	580	620	650	860

Part 6 Medium frequency equipment installation instructions

一、 **Mechanical part installation**

The installation of this part includes the installation of furnace body, dumping furnace electrical, operating table, and water system. The installation must be performed in the following order.

1. General rules of installation

1.1 After the equipment is in place according to the layout plan provided by the design institute or manufacturer, adjust the level and size to meet the requirements of the relevant drawings, then hang the ground bolts, pour cement, and tighten the ground bolts after curing.

1.2 After the furnace body, hydraulic device and console are installed, connect the external hydraulic lines.

1.3 Make the pipeline connection between the total inlet and outlet pipes and the plant water source.

1.4 The connection of each furnace inlet and outlet pipes refer to the water system diagram, in principle, each branch should be equipped with a ball valve. So that each branch circuit is relatively independent and the flow rate can be adjusted.

1.5 Connect the grounding wire of the furnace body, requiring grounding resistance $<4\Omega$.

1.6 Connection of water and oil lines between equipment

2. Connection of water system: The connection of water system includes the connection of primary circulating water and secondary circulating water.

2.1 Primary circulating water (industrial softened water) refers to the cooling water of furnace body, water-cooled cable, capacitor and water-cooled copper busbar, whose installation should be carried out according to the water system diagram and piping layout diagram provided by the design institute or equipment manufacturer.

2.2 Secondary circulating water (pure water) refers to the cooling water of KGPS IF power supply (including rectifier thyristor, DC flat wave reactor, inverter thyristor, inductor and phase change inductor), whose installation should be constructed according to the water system diagram and piping layout diagram provided by the design institute or equipment manufacturer. All piping accessories of the secondary circulating water system must be made of stainless steel (1Cr18Ni9Ti).

2.3 After the water system is connected, do water pressure test, test pressure 0.4Mpa, 15 minutes shall not have leakage.

2.4 Once the installation of the circulating water system should be done after the external cycle test for 4 hours, the cycle test should be every 20 minutes to knock the pipe wall, joints once, so that the oxidation skin, welding slag and debris in the pipeline fully exhausted to keep the pipeline clean and smooth.

Installation of electrical parts

The installation of the electrical part includes the installation of the high-voltage cabinet (or high-voltage load switch), rectifier transformer, KGPS IF power cabinet, RFM compensation capacitor cabinet (including the furnace change switch), and low-voltage power cabinet. The installation must be carried out in the following order.

1. **Equipment in place**.

The equipment should be positioned according to the layout plan provided by the design institute or manufacturer, and adjusted to the level and size to meet the requirements of the relevant drawings.

2. **Equipment fixation**.

2.1 The rectifier transformer should be accurately placed on the foundation track and fixed.

2.2 Other electrical equipment in place, the need for secondary pouring, and then hanging a good footing bolts, pouring cement, and tighten the footing bolts after permaculture; do not need secondary pouring, in the footing hole to play expansion bolts to fix.

3. Connection of the main electric between the equipment.

3.1 The connection between high-voltage 10KV and high-voltage switchgear (or high-voltage load switch) is made by high-voltage cable, which is led to high-voltage switchgear (or high-voltage load switch) from the high-voltage distribution room of the factory overhead or along the high-voltage cable trench; the connection between high-voltage switchgear (or high-voltage load switch) and rectifier transformer is made by high-voltage cable or copper busbar of the same cross-section, and the cable (copper busbar) should be fixed.

3.2 The connection between the rectifier transformer and KGPS IF power supply, KGPS IF power supply cabinet and RFM compensation capacitor cabinet uses overhead copper busbar, and the copper busbar should be fixed.

3.3 The connection between the compensation capacitor cabinet and the water-cooled cable of the furnace body should be overhead water-cooled copper busbar because the current is quite large, and the distance between the water-cooled copper busbar is "80mm, in order to reduce the influence of the inductance of the channel on the IF power supply start; the water-cooled copper busbar should be fixed at 1500mm apart, and the metal part of the fixed bracket should be far away from the copper busbar (d>200mm), in order to reduce the induction heating of the metal body produced by the IF. Induction heating.

3.4 The selection principle of the main electrical coupling fastening bolts.

a) Power frequency 50Hz or DC part of the choice of copper bolts (or galvanized steel bolts)

b) Power frequency 500Hz part of the choice of copper bolts (non-magnetic stainless steel bolts 1Cr18Ni9Ti)

3.5 The principle of color selection for the main electrical copper busbar color coded paint.

a) Three-phase power supply A-phase ---- yellow; B-phase ---- green; C-phase -red

b) IF power supply ---- black

4. Connection of control electrics between equipment.

4.1 Distribution cables

a) The connection between the workshop power distribution room and the low-voltage power cabinet adopts three-phase four-wire power cable with a capacity of 30KVA

b) Low-voltage power cabinet is divided to dumping furnace electrical, water cooling system, medium frequency power using three-phase four-wire cable.

4.2 Control cables

The control cable used for electrical signal contact between equipment should be multi-core control cable, all control cables are buried in the cable trench or wall arrangement through the threading pipe. The threading tubes are fixed at 1m intervals.

5 the inspection before tying the furnace liner.

5.1 Tilting furnace system check: motor rotation direction is correct or not, sound is normal, turn action check, whether the action is flexible, smooth, whether there is jamming, shaking and joint leakage phenomenon, if there is a problem, repair to a good state, the above action shall be carried out back and forth several times until completely normal. It should be noted that due to the cold weather and new seals, there may be a squeaking sound in the tilting furnace test is normal.

5.2 Furnace body inspection.

Check whether the induction paint insulation layer is scratched and damaged during transportation and installation, and repair any damage found.

5.3 Insulation inspection.

Before the inductor is connected to water, check the inductor insulation and voltage strength to ground. HAfter removing all rubber tubes from the inductor and making the inductor fully dry, measure the insulation resistance of the "inductor to ground (furnace body)" with a megohmmeter to be not less than 1MΩ. 1000" pay voltage insulation withstand test for 1 minute, no flashover and breakdown phenomenon.

5.4 Water cooling system inspection.

After the water line is connected, with 0.4Mpa water pressure test for 10 minutes, check whether there is leakage. After passing the electric contact pressure gauge to 0.15 ~ 0.25Mpa range. Set the signal temperature meter to 55°C. Check whether the water flow branch is normal.

5.5 Inducer no-load power feed check.

This work can be done only after the above checks are passed. Under the condition that the furnace is not put into the furnace charge or crucible mold, send power to the inductor to check whether the electrical switch action, line vibration and heating, instrument indication, electrical control line action and the vibration of the inductor itself, and the condition of inter-turn insulation are normal. For the sake of safety, the voltage should be set from low to high in 3~5 steps until the rated voltage is continuously fed for about 30 minutes. Note that the inductor should be connected to water at this time.

Part 7 Medium Frequency Equipment Safety Operation Precautions

Medium frequency induction melting furnace is high temperature electric equipment, safety is of paramount importance, strict operating procedures and handover records should be developed, and the necessary safety measures should be adopted.

1, the iron must not be over-tempered, otherwise, the furnace lining life will drop sharply.

2, in order to prevent mechanical damage to the furnace lining, empty furnace charging should be added to the small fibrous material first, and then large material. When adding large material, must gently join, not to throw, throw, so as not to damage the lining, causing leakage, accidents; at the same time try to avoid the hot state empty furnace case full of cold material, causing cracks in the lining.

3. Remaining iron melting, not only high efficiency, but also high furnace lining life.

4、In order to prevent iron splash, do not add wet and closed cavity furnace material.

5、Prevent the iron surface from "crusting". The cause is too low temperature of the molten iron before charging or too much charging, the material surface is too large (fibrous, crumb material, etc.). Once found "crust", the furnace should be immediately tilted up about 20 °, continue to send electricity to open the "crust" front.

6, "freezing furnace", for some reason (crust does not melt, sudden power failure) the whole furnace iron solidification, the need to remove the lining. Such as a small amount of solidification at the bottom of the furnace, depending on the situation can not be removed from the liner, after the call melted away, but the lining is not good.

7, emergency measures: emergency water or emergency power (power, lighting) or emergency hydraulic pump, the user according to the scale of production and power supply quality to be considered.

8, the furnace is directly below to have a furnace accident pit, laying refractory bricks, once the leakage of the furnace, the iron will quickly flow to the furnace pit. At the same time, the rear side of the furnace oil pipe should be shielded to protect.

9, 9 regularly check the cooling pipeline connections for leakage and poor circulation, especially the medium frequency power supply cooling part, so as not to cause damage to the thyristor.

10、When you need to cut off the whole power supply, you must first cut off the IF cabinet, and then cut off the power supply of the low-voltage cabinet.

11、Safety facilities: furnace basement and electrical room according to the charged, a large amount of oil exists in the working conditions, should be used respectively different fire apparatus. Furnace workers, electrical operators must learn to use the above utensils to ensure the timely and rapid elimination of accidents, workers should have protective equipment; electrical room operating parts of the floor to lay insulated rubber sheet, easy to electrocution parts with protective cover, etc.

Part 8 MF equipment maintenance and precautions

After the furnace sintering is completed, it can be put into normal operation. The most problematic furnace operation is mostly in the lining life, requiring operators to be responsible, strict and meticulous, always pay attention to check the lining work and keep the whole system in good condition.

1、The dumping furnace system should be injected with lubricant regularly.

2, cooling water temperature rise condition, when using tap water should be regularly checked coil, cable internal scaling, when the scale of 0.5mm or more should be 5% of HCl cycle cleaning.

3、Check regularly whether the furnace body grounding is normal <4Ω.

4, often check the degree of furnace lining erosion, to achieve timely repair, in order to extend the life of the furnace lining, to achieve economic operation.

5, when removing the furnace lining, in order to protect the coil insulation, the steel brazier can not hit the lining vertically, it is best to remove the lining at the left (or right) side of the furnace mouth 300mm from top to bottom, and then expand outward. Many units due to the incorrect way to remove the lining, resulting in damage to the coil and even deformation and scrap.

6、During operation or when the crucible temperature is still high when the furnace is empty, the number of times and time of opening the furnace cover should be reduced to reduce the heat loss and the rapid cooling crack of the furnace lining.

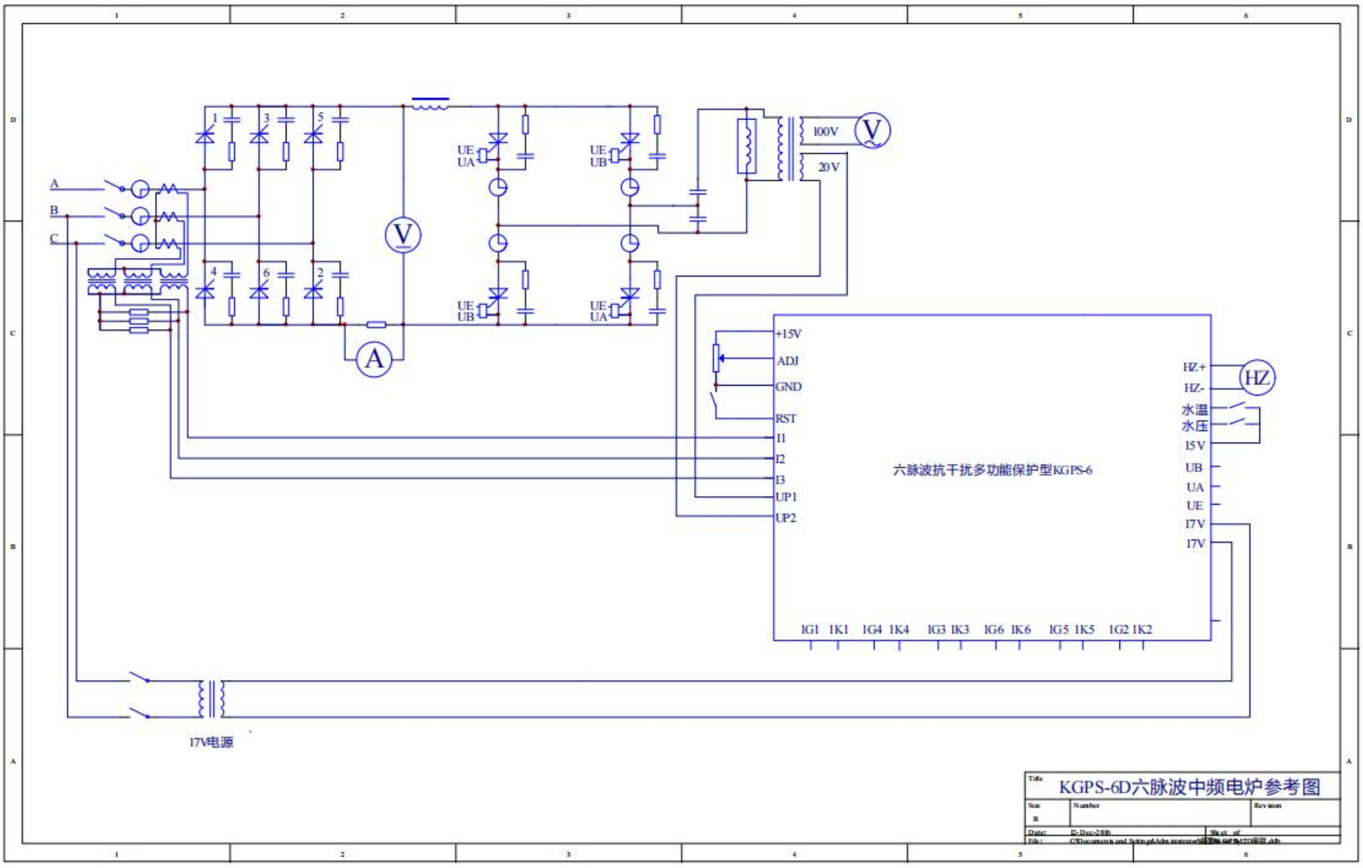
7、The rotating parts of the equipment should be regularly lubricated with sufficient oil.

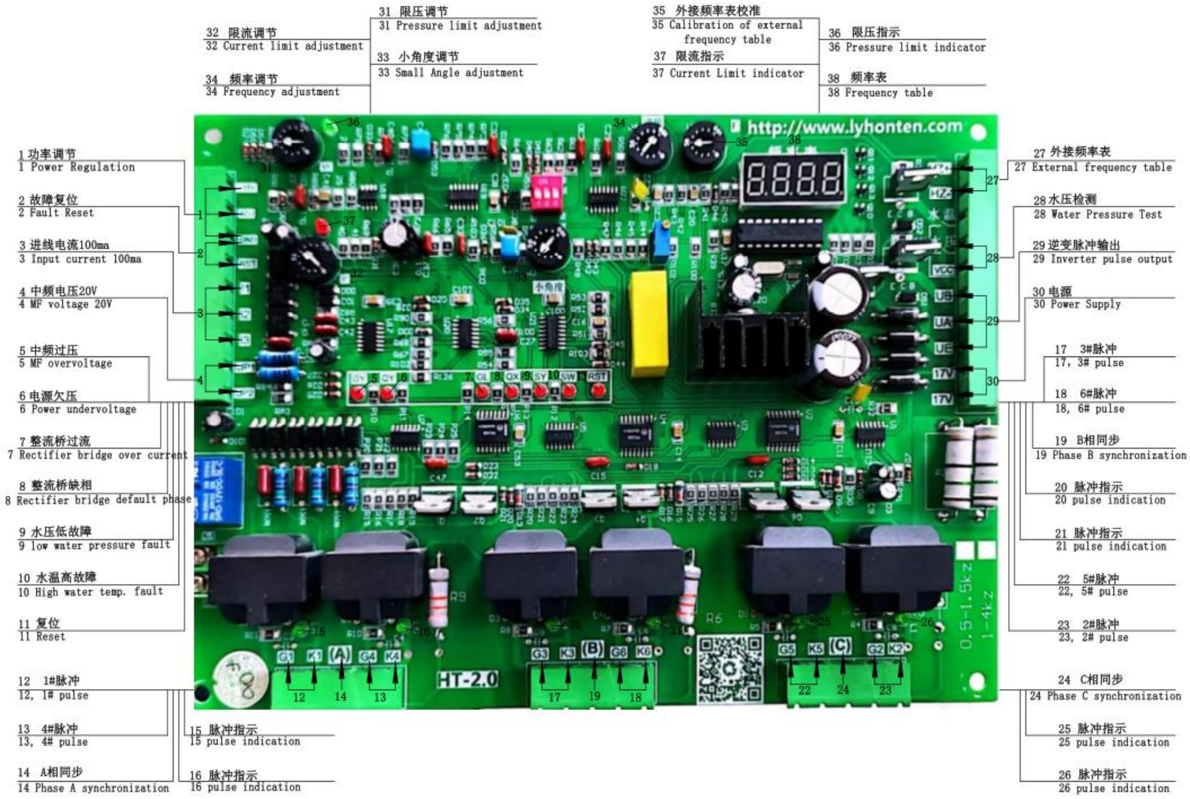
8、Check frequently whether the bolts at each joint of inductor, cable, converging copper row and electrical room bus bar are loose to ensure that the conductive parts are in good contact, and dust regularly to keep clean.

9、Check the tightening of the induction coil frequently to prevent loosening.

10、If the furnace stops using for a longer period of time, the memory cooling water should be blown out to prevent the induction coils and pipes from rusting (in the cold season also to prevent freezing and cracking the coils and pipes).

Part 9, Attached




32 限流调节
32 Current limit adjustment
31 限压调节
31 Pressure limit adjustment
33 小角度调节
33 Small Angle adjustment
34 频率调节
34 Frequency adjustment
35 外接频率表校准
35 Calibration of external frequency table
37 限流指示
37 Current Limit indicator
36 限压指示
36 Pressure limit indicator
38 频率表
38 Frequency table
1 功率调节
1 Power Regulation
2 故障复位
2 Fault Reset
3 进线电流100ma
3 Input current 100ma
4 中频电压20V
4 MF voltage 20V
5 中频过压
5 MF overvoltage
6 电源欠压
6 Power undervoltage
7 整流桥过流
7 Rectifier bridge over current
8 整流桥缺相
8 Rectifier bridge default phase
9 水压低故障
9 low water pressure fault
10 水温高故障
10 High water temp. fault
11 复位
11 Reset
12 1#脉冲
12, 1# pulse
13 4#脉冲
13, 4# pulse
14 A相同步
14 Phase A synchronization
15 脉冲指示
15 pulse indication
16 脉冲指示
16 pulse indication
27 外接频率表
27 External frequency table
28 水压检测
28 Water Pressure Test
29 逆变脉冲输出
29 Inverter pulse output
30 电源
30 Power Supply
17 3#脉冲
17, 3# pulse
18 6#脉冲
18, 6# pulse
19 B相同步
19 Phase B synchronization
20 脉冲指示
20 pulse indication
21 脉冲指示
21 pulse indication
22 5#脉冲
22, 5# pulse
23 2#脉冲
23, 2# pulse
24 C相同步
24 Phase C synchronization
25 脉冲指示
25 pulse indication
26 脉冲指示
26 pulse indication
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HT-2.0